

Gendered Scrutiny under Scarcity: First Impressions, Endogenous Information Search, and a Novel Economic Conjoint Framework for Electoral Judgment

April 27, 2026

Introduction

Research in political psychology and political behavior has long shown that voters rely on low-information cues when evaluating candidates (Lau and Redlawsk 2001), especially facial appearance (D. Dion 1998; Stockemer and Praino 2019), perceived competence (Herrmann and Shikano 2016), attractiveness (K. Dion, Berscheid, and Walster 1972; Rosar, Klein, and Beckers 2008), trustworthiness, and other first-impression signals (Bahamonde and Sarpila 2024). At the same time, work in political economy and behavioral decision-making suggests that information use is shaped not only by limited cognition (Redlawsk 2001) but also by the costs of acquiring and processing additional evidence (Zaller 1992). Yet these literatures have rarely been integrated in a way that treats information search itself as part of the political judgment process (but see Lau and Redlawsk (2006)). Most existing studies ask whether appearance affects evaluations (Berggren, Jordahl, and Poutvaara 2010, 2017), or whether providing new information changes preferences (Mattes et al. 2010; Todorov et al. 2005). Much less is known about whether voters actively *choose* to remain at the level of first impressions, when they decide to investigate further, and whether those search decisions are themselves structured by gender.

This gap is especially important because first impressions appear to emerge extremely quickly and to persist even when additional information is available. The literature on candidate evaluation consistently suggests that photographs allow voters to form immediate impressions about candidates (Banducci et al. 2008). Related work shows that such impressions can arise under extremely short exposure times (e.g., 100 milliseconds) and are often only weakly corrected by subsequent information (Mattes et al. 2010; Todorov et al. 2005). Antonakis and Dalgas (2009) even report that children have the same (and sometimes even *better*) predicting skills when choosing winning political candidates. **If so, the key question is not simply whether facial appearance matters, but under what conditions respondents choose to verify, override, or act on those initial inferences.**

This paper argues that electoral judgment is not only a matter of which candidate traits

voters prefer, but also of how much effort they are willing to expend to learn about different candidates. That question is especially important in the case of women in politics. A large literature suggests that female candidates are often evaluated through gendered expectations and may face distinct standards of competence, likability, and suitability for office. But the existing evidence focuses mainly on vote choice, candidate evaluations, or stereotype activation (Ditonto and Mattes 2018; Krook and Restrepo Sanín 2020). It tells us whether gender matters for outcomes, but not whether *gender shapes the process of attention allocation itself*. We contend that one of the key mechanisms of unequal political judgment may lie precisely there: **female candidates may elicit greater information demand because respondents feel a stronger need to verify, resolve, or reconcile their initial impressions**. In that sense, political judgment is distributed unevenly across candidates.

Our core theoretical claim is that first impressions generate different levels of subjective confidence, and that this confidence governs the demand for additional information. When respondents believe that a candidate’s face already provides a sufficiently clear signal, they may rationally decline to spend scarce resources on further investigation. Low search may therefore reflect efficient economizing if first impressions are informative and may also reflect overconfidence if respondents rely too heavily on facial heuristics. The gendered implication is especially important: if female candidates are judged under stricter priors (Ditonto and Mattes 2018; Krook and Restrepo Sanín 2020), respondents may be less willing to trust first impressions alone and more likely to invest in additional search. The result is a stronger demand for information not because women inherently provide less information, but because initial impressions of female candidates may activate motivated reasoning (Kunda 1990), inducing respondents to seek and evaluate additional evidence under a higher burden of verification.

To study this process, we introduce a novel *economic conjoint experiment* implemented in oTree (see prototype here). The design combines the multidimensional choice structure of conjoint analysis with an explicitly economic search environment in which information is scarce, costly, and endogenously acquired. Rather than presenting respondents with a fully revealed bundle of candidate attributes, we allow them to uncover information selectively by spending points from a limited budget. This transforms the conjoint from a passive stated-preference task into a sequential decision problem in which *search behavior* itself becomes theoretically meaningful. Respondents first encounter real candidate photographs in a low-information setting, form an initial judgment, and then decide whether to pay to reveal additional profile attributes before updating their choice (see Figure 1). The experiment therefore identifies not only which candidate features affect electoral judgments, but also which features respondents consider worth learning about, under what conditions, and for whom.

The design also incorporates task-level time pressure. In some choice tasks assigned at random, respondents face a visible timer and a reward structure in which delay reduces the value of a correct answer. The purpose of this feature is to isolate the conditions under which respondents rely more heavily on first impressions when deliberation becomes costly. If rapid perceptual inferences

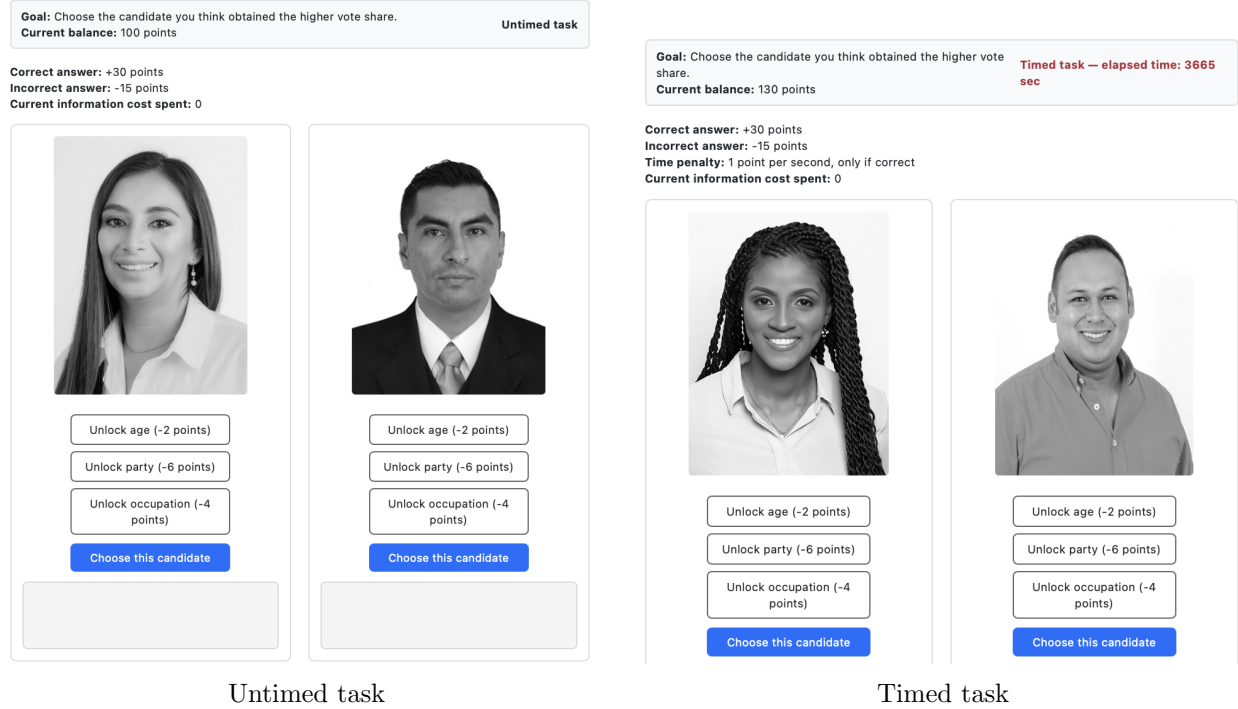


Figure 1: Prototype screens of the economic conjoint experiment

are indeed powerful and persistent, then timed tasks provide a direct way to examine whether respondents fall back on facial heuristics when the opportunity to reflect is constrained.

This methodological contribution is substantial. Standard conjoint experiments are powerful tools for estimating the causal effects of profile attributes on selection probabilities (Hainmueller, Hopkins, and Yamamoto 2014; Leeper, Hobolt, and Tilley 2020), but they typically reveal much more about respondents’ final choices than about the information-processing strategies that produced them. Recent work using eye-tracking shows that respondents in conjoint experiments selectively attend to some attributes and profiles while ignoring others, in ways consistent with bounded rationality and cognitive cost minimization (Jenke et al. 2021). Our economic conjoint experiment extends this insight by making information acquisition itself endogenous and costly. **Rather than inferring information use from researcher-imposed, fully revealed profiles, we design an environment in which respondents must decide for themselves whether additional information is worth acquiring before making a decision.**

We expect respondents to spend more points when visual cues are ambiguous or incongruent. We also expect this logic to be gendered: **female candidates should trigger greater information demand**, consistent with a politics of asymmetric scrutiny in which women are evaluated under a higher burden of verification.

Technical Methodological Addendum

Core design features

- **Unit of evaluation.** Each respondent $i = 1, \dots, N$ completes T repeated choice tasks. In each task $t = 1, \dots, T$, the respondent sees two candidate profiles, indexed by $j \in \{L, R\}$.
- **Visual first exposure.** Each profile is initially presented only through the candidate's photograph. This creates a low-information stage in which respondents must first rely on first impressions before deciding whether to acquire additional information.
- **Endogenous information acquisition.** For each profile, respondents can selectively unlock three attributes: party, age, and occupation. Let $k \in \{\text{party, age, occupation}\}$ index the attribute. Unlocking is costly and therefore behaviorally meaningful.
- **Randomization of candidate exposure.** Candidate photographs are randomly drawn from the candidate pool and then randomly assigned to the left or right side of the task screen. This randomization ensures that any side-specific effect is orthogonal to candidate characteristics.
- **Randomization of information order.** The vertical order of the three unlockable attributes is randomized at the respondent level and then held fixed across tasks for that respondent. This prevents systematic ordering effects while preserving within-respondent comparability.
- **Randomization of time pressure.** Time pressure is randomized at the task level. A given task may or may not include the timer, so respondents can experience untimed tasks only, timed tasks only, or a mixture of both across the survey.
- **Budget-constrained search.** Each respondent receives an initial endowment and must manage a running point balance across tasks. Correct choices increase the balance, incorrect choices reduce it, and each information unlock carries an explicit cost. In timed tasks, delay further reduces the net payoff from a correct choice.
- **Behavioral measurement.** The design records which attributes are unlocked, in which task, for which profile, how much is spent on information, how much time is used, whether the respondent was correct, and the respondent's balance after each task.
- **Substantive outcome.** In the current implementation, respondents are asked to identify which candidate obtained the higher vote share. This aligns the forced choice with an externally observable electoral outcome rather than with an unobserved personal preference.

Main quantities of interest

The central advantage of the design is that it yields causal quantities not only for final choices, but also for the *search process* itself. Let $G_{ijt} \in \{0, 1\}$ denote whether profile j shown to respondent i in task t is female, and let $S_{it} \in \{0, 1\}$ denote whether task t is timed.

Let

$$U_{ijkt} \in \{0, 1\}$$

indicate whether respondent i unlocks attribute k for profile j in task t .

The first quantity of interest is the average effect of candidate gender on the probability that a given attribute is unlocked:

$$\tau_k = \mathbb{E}[U_{ijkt}(1, 0) - U_{ijkt}(0, 0)].$$

Here, τ_k captures whether female candidates generate greater demand for information on attribute k in the absence of time pressure. If the theory of asymmetric scrutiny is correct, we expect $\tau_k > 0$ for at least some attributes.

A second quantity of interest is the effect of time pressure on information acquisition:

$$\delta_k = \mathbb{E}[U_{ijkt}(0, 1) - U_{ijkt}(0, 0)].$$

This estimand captures whether respondents become less likely to acquire costly information when decision time is made more salient.

The key interaction quantity of interest is:

$$\gamma_k = \mathbb{E}[(U_{ijkt}(1, 1) - U_{ijkt}(0, 1)) - (U_{ijkt}(1, 0) - U_{ijkt}(0, 0))].$$

This difference-in-differences estimand captures whether the gender gap in information demand changes under time pressure. It is the most direct formalization of the claim that female candidates may face a higher burden of verification, especially when respondents are forced to make decisions under scarcity.

Because the design also records the total amount spent on information, a second family of quantities of interest concerns search intensity. Let

$$C_k > 0$$

denote the point cost of unlocking attribute k . Define total information expenditure for respondent i in task t as

$$E_{it} = \sum_{j \in \{L, R\}} \sum_k C_k U_{ijkt}.$$

Then the average causal effect of candidate gender on total search expenditure can be written as

$$\tau_E = \mathbb{E}[E_{it}(1, 0) - E_{it}(0, 0)] ,$$

with the obvious timed analogue and interaction extension.

The design also permits causal analysis of final task performance. Let

$$Y_{it} \in \{0, 1\}$$

indicate whether respondent i correctly identifies the candidate with the higher vote share in task t . Then one can define:

$$\tau_Y = \mathbb{E}[Y_{it}(1, 0) - Y_{it}(0, 0)] ,$$

to study whether the presence of a female candidate affects prediction accuracy, and

$$\gamma_Y = \mathbb{E}[(Y_{it}(1, 1) - Y_{it}(0, 1)) - (Y_{it}(1, 0) - Y_{it}(0, 0))] ,$$

to assess whether time pressure differentially changes accuracy when female candidates are present.

Number of tasks

For the present study, I recommend using

$$T = 8$$

choice tasks per respondent.

This choice is methodologically conservative. It sits comfortably within the logic of Bansak et al. (2018), who find little evidence of substantial deterioration in conjoint responses even with much larger numbers of tasks. Using eight tasks therefore provides a reasonable compromise between statistical power and respondent burden, especially since the present design is cognitively more demanding than a standard fully revealed conjoint because respondents must actively manage costly information acquisition.

Back-of-the-envelope sample size calculation

The main power calculation should be anchored in the candidate-level information acquisition outcome U_{ijkt} , since the proposal's central theoretical claim concerns differential *information demand*. Suppose the target effect size is a five-percentage-point difference in the probability of unlocking a given attribute, for example:

$$\Pr(U_{ijkt} = 1 \mid G_{ijt} = 1) - \Pr(U_{ijkt} = 1 \mid G_{ijt} = 0) = 0.05.$$

Using a conventional two-sided test with $\alpha = 0.05$ and power 0.80, and taking a baseline

unlock probability of roughly 0.30, the required independent sample size is approximately

$$n_{\text{indep}} \approx 1,376$$

profile observations per arm.

In the present design, however, observations are clustered within respondents because each respondent evaluates multiple profiles. With $T = 8$ tasks and two profiles per task, each respondent contributes up to

$$m = 2T = 16$$

candidate-profile exposures. A standard design-effect correction gives:

$$DE = 1 + (m - 1)\rho,$$

where ρ is the within-respondent intraclass correlation in profile-level outcomes.

For plausible values such as $\rho \in [0.10, 0.20]$, this yields:

$$DE \in [2.5, 4.0].$$

The effective number of profile exposures per arm is therefore approximately

$$n_{\text{eff, arm}} \approx \frac{8N}{DE},$$

because, under balanced randomization, each respondent contributes on average about $T = 8$ profile exposures to each gender arm.

Setting

$$\frac{8N}{DE} \geq 1,376$$

implies:

$$N \geq \frac{1,376 \times DE}{8}.$$

This gives a required respondent sample approximately in the range:

$$N \in [430, 688].$$

That range is appropriate for the *main effect* of candidate gender on attribute unlocking. Because the project is also interested in interactions with time pressure and in heterogeneity across multiple unlockable attributes, a more conservative target is warranted. I therefore recommend a target sample of:

$$N = 1,000$$

respondents, with

$$T = 8$$

tasks each.

This design would generate:

$$2NT = 16,000$$

candidate-profile exposures in total. Under balanced assignment, this yields substantial information for estimating both main effects and gender-by-time-pressure interactions while remaining comfortably within the range of task counts that Bansak et al. (2018) suggest are unlikely to create severe satisficing problems.

Relation to AMCEs, Marginal Means, and Framing Designs

It is useful to distinguish clearly between the quantities of interest that dominate the standard conjoint framework and the quantities of interest targeted by the present design. In the Hainmueller, Hopkins, and Yamamoto framework, the principal causal estimand is the average marginal component effect (AMCE), which summarizes the effect of changing one attribute level on the probability that a profile is chosen, averaging over the joint distribution of the remaining attributes. In their notation, if $Y_{ijk}(t)$ denotes the potential choice outcome for respondent i in task k when profile vector t is shown, and if $T_{ijk\ell}$ denotes the ℓ th attribute of the profile, then the AMCE for attribute ℓ compares two levels, say x_1 and x_0 , averaging over all possible realizations of the remaining profile components. Substantively, the AMCE answers the question: by how much does support for a profile change when one attribute is shifted from x_0 to x_1 , holding fixed the fully revealed conjoint structure and averaging over the distribution of the other attributes?

Leeper, Hobolt, and Tilley show that AMCEs are closely tied to marginal means. In a fully randomized forced-choice conjoint, a marginal mean for feature level x can be written as

$$MM_\ell(x) = \Pr(Y_{ijk} = 1 \mid T_{ijk\ell} = x),$$

where the probability is taken over the randomized distribution of all remaining profile attributes. In fully randomized designs, the AMCE is simply the difference between two such marginal means:

$$AMCE_\ell(x_1, x_0) = MM_\ell(x_1) - MM_\ell(x_0).$$

Thus, AMCEs summarize relative differences against a reference level, whereas marginal means summarize the absolute favorability of profiles carrying a given feature level. In both cases, however, the estimands are defined over *fully revealed* profile content and linked to the respondent’s final choice.

The present design differs because the central behavioral question arises *before* the final choice is made. Respondents do not receive the full profile ex ante. Instead, they first observe the candidate’s photograph and then decide whether to unlock party, age, and occupation under a binding budget and, in some tasks, under time pressure. As a result, the primary quantities of interest are not initially AMCEs of fully displayed attributes on the final choice outcome Y_{ikt} , but

quantities governing *information acquisition itself*. Let

$$U_{ijktm} \in \{0, 1\}$$

indicate whether respondent i unlocks attribute $m \in \{\text{party, age, occupation}\}$ for profile j in task k . Let $G_{ijk} \in \{0, 1\}$ indicate whether the profile is female and let $S_{ik} \in \{0, 1\}$ indicate whether task k is timed. Then the main estimands of the present design are quantities such as

$$\tau_m = \mathbb{E}[U_{ijktm}(1, 0) - U_{ijktm}(0, 0)],$$

which captures the effect of candidate gender on the probability that attribute m is unlocked, and

$$\gamma_m = \mathbb{E}[(U_{ijktm}(1, 1) - U_{ijktm}(0, 1)) - (U_{ijktm}(1, 0) - U_{ijktm}(0, 0))],$$

which captures whether time pressure changes the gender gap in information demand.

These estimands do not contradict AMCEs or marginal means. Rather, they operate at a different stage of the decision process. AMCEs and marginal means ask: *once an attribute is part of the revealed profile, how does it shape support?* The quantities of interest in the present design instead ask: *is that attribute acquired at all, under what conditions, and for whom?* In this sense, the current design complements the standard conjoint framework by shifting attention from the causal effects of fully revealed attributes on final choice to the causal structure of selective information acquisition under scarcity. That is precisely the dimension that standard fully revealed conjoint designs are not built to capture.

This distinction also clarifies the limits of standard AMCE and marginal-mean analyses for the present application. If one were to compute an AMCE of party, age, or occupation on final choice in the usual way, one would be analyzing the consequences of those attributes *conditional on exposure*. But in the present design, exposure is itself endogenous: respondents must choose whether to incur a cost in order to observe the attribute. For that reason, standard AMCEs and marginal means cannot by themselves recover the key behavioral object of interest, namely the process by which some pieces of information become visible while others remain unseen. At most, they can describe how information matters *after it has entered the respondent's choice set*. The present design is therefore best understood as extending, rather than replacing, the conjoint framework: it preserves the multidimensional profile-comparison logic of conjoint analysis while introducing a prior stage in which information acquisition itself becomes a theoretically meaningful outcome.

It is equally important to clarify that this is not a standard framing experiment. In a conventional framing design, respondents are typically assigned to one of a small number of externally imposed messages or interpretive presentations, and the central estimand compares outcomes across those frame assignments. Formally, if $F_i \in \{0, 1\}$ indexes exposure to two alternative frames and

$Y_i(F_i)$ denotes the corresponding potential outcome, the canonical framing estimand is

$$\Delta_F = \mathbb{E}[Y_i(1) - Y_i(0)].$$

The frame is exogenous, uniform within condition, and delivered directly by the researcher. The inferential target is the causal effect of *how the same object is presented*. By contrast, the present design does not assign respondents to a single message frame that exogenously alters interpretation. Instead, it assigns respondents to a decision environment in which multiple profile components may or may not become visible depending on respondents' own acquisition choices. What varies is not simply the wording or presentation of a fixed object, but the respondent's self-directed exposure to profile information under cost and time constraints.

Using notation, a framing experiment takes the form

$$F_i \longrightarrow Y_i,$$

whereas a standard fully revealed conjoint takes the form

$$T_{ijk} \longrightarrow Y_{ijk},$$

with T_{ijk} denoting the vector of randomized profile attributes. The present design instead has the structure

$$\text{image}_{ijk} \longrightarrow U_{ijktm} \longrightarrow Y_{ik},$$

possibly moderated by task-level time pressure S_{ik} . The crucial difference is that the information set entering the final judgment is not wholly assigned by the researcher at the moment of treatment delivery. It is jointly produced by the experimental environment and the respondent's own acquisition decisions. For that reason, the design is not well described as a framing experiment, even though it manipulates the informational environment. Its distinctive feature is not exogenously imposed interpretation, but endogenous, costly, and sequential revelation of profile content.

A useful way to summarize the comparison is therefore as follows. In the standard conjoint framework,

$$T_{ijk} \longrightarrow Y_{ijk},$$

and the inferential task is to estimate how the randomized components of a fully revealed profile affect the final choice. In a framing design,

$$F_i \longrightarrow Y_i,$$

and the inferential task is to estimate how alternative presentations of an object affect judgment. In the present framework, however,

$$\text{image}_{ijk} \longrightarrow U_{ijktm} \longrightarrow Y_{ik},$$

with the intermediate stage U_{ijktm} itself treated as a substantive outcome. The methodological contribution of the present design lies precisely in opening this intermediate stage to direct analysis. Standard AMCEs and marginal means remain useful once attributes are observed, and framing logic remains relevant for thinking about informational environments, but neither framework by itself captures a design in which respondents must first decide whether additional information is worth acquiring.

References

- Antonakis, John and Olaf Dalgas (2009). “Predicting Elections: Child’s Play!” In: *Science* 323.5918, pp. 1183–1183. ISSN: 0036-8075. DOI: 10.1126/science.1167748.
- Bahamonde, Hector and Outi Sarpila (June 2024). “Physical appearance and elections: An inequality perspective”. In: *Political Psychology* 45.3, pp. 623–642. ISSN: 0162-895X. DOI: 10.1111/pops.12940. URL: <https://onlinelibrary.wiley.com/doi/10.1111/pops.12940>.
- Banducci, Susan et al. (Oct. 2008). “Ballot Photographs as Cues in Low-Information Elections”. In: *Political Psychology* 29.6, pp. 903–917. ISSN: 0162895X. DOI: 10.1111/j.1467-9221.2008.00672.x. URL: <https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9221.2008.00672.x>.
- Berggren, Niclas, Henrik Jordahl, and Panu Poutvaara (Feb. 2010). “The Looks of a Winner: Beauty and Electoral Success”. In: *Journal of Public Economics* 94.1-2, pp. 8–15. ISSN: 00472727. DOI: 10.1016/j.jpubeco.2009.11.002. URL: <http://dx.doi.org/10.1016/j.jpubeco.2009.11.002><https://linkinghub.elsevier.com/retrieve/pii/S004727270900142X>.
- (2017). “The right look: Conservative politicians look better and voters reward it”. In: *Journal of Public Economics* 146, pp. 86–103. DOI: 10.1016/j.jpubeco.2016.12.008.
- Dion, Douglas (Jan. 1998). “Evidence and Inference in the Comparative Case Study”. In: *Comparative Politics* 30.2, p. 127. ISSN: 00104159. DOI: 10.2307/422284. URL: <https://www.jstor.org/stable/422284?origin=crossref>.
- Dion, Karen, Ellen Berscheid, and Elaine Walster (1972). “What is Beautiful is Good”. In: *Journal of Personality and Social Psychology* 24.3, pp. 285–290. ISSN: 1939-1315. DOI: 10.1037/h0033731.
- Ditonto, Tessa and Kyle Mattes (2018). “Differences in Appearance-Based Trait Inferences for Male and Female Political Candidates”. In: *Journal of Women, Politics & Policy* 39.4, pp. 430–450. ISSN: 1554-477X. DOI: 10.1080/1554477X.2018.1506206.
- Hainmueller, Jens, Daniel J. Hopkins, and Teppei Yamamoto (2014). “Causal inference in conjoint analysis: Understanding multidimensional choices via stated preference experiments”. In: *Political Analysis* 22.1, pp. 1–30. DOI: 10.1093/pan/mpt024.
- Herrmann, Michael and Susumu Shikano (June 2016). “Attractiveness and Facial Competence Bias Face-Based Inferences of Candidate Ideology”. In: *Political Psychology* 37.3, pp. 401–417. ISSN: 0162895X. DOI: 10.1111/pops.12256. URL: <https://onlinelibrary.wiley.com/doi/10.1111/pops.12256>.
- Jenke, Libby et al. (Jan. 2021). “Using Eye-Tracking to Understand Decision-Making in Conjoint Experiments”. In: *Political Analysis* 29.1, pp. 75–101. ISSN: 1047-1987. DOI: 10.1017/pan.2020.11. URL: https://www.cambridge.org/core/product/identifier/S104719872000011X/type/journal%7B%5C_%7Darticle.
- Krook, Mona Lena and Juliana Restrepo Sanín (Sept. 2020). “The Cost of Doing Politics? Analyzing Violence and Harassment against Female Politicians”. In: *Perspectives on Politics* 18.3, pp. 740–

755. ISSN: 1537-5927. DOI: 10.1017/S1537592719001397. URL: https://www.cambridge.org/core/product/identifier/S1537592719001397/type/journal%7B%5C_%7Darticle.
- Kunda, Ziva (Nov. 1990). "The Case for Motivated Reasoning". In: *Psychological bulletin* 108.3, pp. 480–498. ISSN: 0033-2909. URL: <http://www.ncbi.nlm.nih.gov/pubmed/2270237>.
- Lau, Richard and David Redlawsk (Oct. 2001). "Advantages and Disadvantages of Cognitive Heuristics in Political Decision Making". In: *American Journal of Political Science* 45.4, p. 951. ISSN: 00925853. DOI: 10.2307/2669334. URL: <https://www.jstor.org/stable/2669334?origin=crossref>.
- (2006). *How Voters Decide, Information Processing in Election Campaigns*. Cambridge University Press.
- Leeper, Thomas, Sara Hobolt, and James Tilley (2020). "Measuring Subgroup Preferences in Conjoint Experiments". In: *Political Analysis* 28.2, pp. 207–221. ISSN: 1047-1987. DOI: 10.1017/pan.2019.30.
- Mattes, Kyle et al. (Feb. 2010). "Predicting Election Outcomes from Positive and Negative Trait Assessments of Candidate Images". In: *Political Psychology* 31.1, pp. 41–58. ISSN: 0162895X. DOI: 10.1111/j.1467-9221.2009.00745.x. URL: <https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9221.2009.00745.x>.
- Redlawsk, David (Feb. 2001). "You Must Remember This: A Test of the On-Line Model of Voting". In: *The Journal of Politics* 63.1, pp. 29–58. ISSN: 0022-3816. DOI: 10.1111/0022-3816.00058. URL: <https://www.journals.uchicago.edu/doi/10.1111/0022-3816.00058>.
- Rosar, Ulrich, Markus Klein, and Tilo Beckers (Sept. 2008). "The Frog Pond Beauty Contest: Physical Attractiveness and Electoral Success of the Constituency Candidates at the North Rhine-Westphalia State Election of 2005". In: *European Journal of Political Research* 47.1, pp. 64–79. ISSN: 0304-4130. DOI: 10.1111/j.1475-6765.2007.00720.x. URL: <https://onlinelibrary.wiley.com/doi/10.1111/j.1475-6765.2007.00720.x>.
- Stockemer, Daniel and Rodrigo Praino (2019). "The Good, the Bad and the Ugly: Do Attractive Politicians Get a 'Break' When They are Involved in Scandals?" In: *Political Behavior* 41.3, pp. 747–767. ISSN: 0190-9320. DOI: 10.1007/s11109-018-9469-1.
- Todorov, Alexander et al. (2005). "Inferences of competence from faces predict election outcomes". In: *Science* 308.5728, pp. 1623–1626. DOI: 10.1126/science.1110589.
- Zaller, John (Aug. 1992). *The Nature and Origins of Mass Opinion*. Cambridge University Press. ISBN: 9780521404495. DOI: 10.1017/CB09780511818691. URL: <https://www.cambridge.org/core/product/identifier/9780511818691/type/book>.